24MT2012 Mathematical Optimization. 9/12/25.

CO1 → Introduction to optimization.
CO2 → Integer programming
CO3 → Non-linear programming
CO4 → Infinite Dimensional Optimization.

CO1 - Introduction to Optimization.

→ Mathematical optimization was firstly used in the problem of allocation scare military resources during 2nd world war. After research of success operation in military field the operation search explored it's ~~exp.~~ existence in industrial areas.

Scope: of mathematical optimization.
→ Used in following fields.

1) agriculture
2) Finance
3) Industry
4) Marketing
5) Producting management.
6) LIC insurance.

Optimization definition :- It is a mathematical problem of finding the best sln from set of feasible solutions.

Objective function :- A func to be maximized or minized

Decision Variables :- The variable that effect outcome.

Constraints :- satisfy conditions.

Types of problems :-

1) Linear progsamming :- obj and constraints are linear.
2) Non-linear progsamming :- At least one non-linear component.
3) Integer progsamming :- Variables restricted to integers.
4) Dynamic programming :- Sequentical decision making.